

Individual Addendum: Olivia Williamson **Name:** Olivia Williamson **Team:** ILOVECODING **Date:** May 1, 2026 --- ## 1. Personal Contribution My contributions to the ILOVECODING capstone project encompassed statistical strategy, difference-in-differences design, final synthesis, and quality assurance across all four milestones with emphasis on econometric rigor and end-to-end validation. - **Milestone 1 (Data Architecture & Documentation):** Designed REIT-FRED merge strategy using year_month keys; created entity identifier mapping table (273 REITs cross-referenced); conducted variable codebook with definitions and units; created data summary statistics table showing coverage, ranges, and missing-value rates. **14 hours** - **Milestone 2 (Statistical Exploration & Hypothesis Formation):** Conducted correlation matrix analysis across 5+ variables; tested normality assumptions (Shapiro-Wilk test: $p < 0.001$ confirming non-normality); identified preliminary leverage-return relationship ($r \approx 0.07$); preliminary beta-return correlation ($r \approx 0.22$); drafted methodology section of M2 report. **13 hours** - **Milestone 3 (DiD Model Design & Implementation):** Designed difference-in-differences specification to test large-cap $\rightarrow$ small-cap heterogeneous effects of 2015 Fed shock; specified treatment variable (market-cap quartile), shock date (2015), and interaction term; conducted parallel-trends visual inspection; estimated Model B; interpreted null result. **14 hours** - **Milestone 4 (Results Interpretation & Executive Communication):** Co-authored Results interpretation section translating regression output into business language (61 bps = 7.3% annualized); drafted sector recommendations with economic rationale (Industrial overweight 15%, Retail underweight 15%); created Conclusions section with specific investment implications; final quality assurance and peer review. **14 hours** **Total Capstone Hours:** ~55 hours --- ## 2. One Defended Methodological Decision *Decision: Employ difference-in-differences identification using firm size (large-cap vs. small-cap) as treatment proxy for 2015 Fed shock, rather than exploiting leverage directly as instrument.* **Reasoning:** In Milestone 3, we faced a fundamental econometric challenge: the leverage-return relationship is potentially endogenous (reverse causality: high-return REITs may increase leverage). We considered two identification strategies: 1. **Direct leverage approach:** Use lagged leverage in fixed effects (our Model A primary specification) 2. **Indirect via DiD:** Use firm size as exogenous proxy for rate sensitivity, test heterogeneous effects post-2015 shock (our Model B) We chose DiD because: - **Size is exogenous to monthly returns.** Market cap cannot be manipulated by REIT managers within a month in response to return realizations. This breaks the reverse-causality concern. - **Economic rationale:** Large REITs typically carry proportionally more variable-rate debt ($p=0.042$ in model shows large-caps earn 27 bps more pre-2015), making them more rate-sensitive. The 2015 Fed shock provides a natural experiment: if rate sensitivity is real, large-cap REITs should underperform post-shock. - **Evidence from M2:** Correlation analysis showed firm size weakly correlates with leverage ($r \approx 0.31$), confirming

size is not perfectly collinear with debt and captures a distinct dimension of risk. The null DiD result ($p=0.247$) is actually informative: it suggests either large REITs hedged the 2015 shock successfully, or size is not the binding dimension of rate exposure. This strengthens rather than weakens our interpretation: the null finding rules out one story (simple leverage-driven returns), pointing us toward systematic risk (beta) as the true driver. --- ## 3. One Key Limitation

***Limitation:** Sample period (2000–2024) includes the 2008 financial crisis, a regime-shift event that may violate fixed effects assumptions of constant parameters.

****Why It Matters:**** Our regression assumes the relationship between leverage and returns is constant over the entire 25-year sample. However, the 2008 crisis was an unprecedented shock: credit markets froze, many structural assumptions about REIT financing broke down, and risk premiums spiked. If the true leverage-return relationship is $\beta_{\text{crisis}} \neq \beta_{\text{normal}}$, then estimating a single constant β over both periods yields a blend of two regime-specific coefficients, reducing interpretability.

****Impact on Findings:****

- ****Beta coefficient is less affected.**** Systematic risk (correlation with market) likely exists in both normal and crisis regimes, just with different magnitudes. Our $t=4.46$ estimates average 2000–2007 effect + 2008–2009 crisis effect + 2010–2024 recovery effect. The "typical" beta premium may be different from our integrated estimate.
- ****Leverage finding is more fragile.**** Leverage effects could differ dramatically pre/post-crisis: pre-2008, REITs could reduce rates via leverage increases; post-2008, REITs faced refinancing constraints. If pre-crisis leverage increased returns but post-crisis it didn't, our overall coefficient is biased toward the sample-weighted average.
- ****Generalization is compromised.**** Forecasting future leverage-return dynamics using 2000–2024 parameters assumes 2008 won't recur. If market structure has changed post-crisis (e.g., higher capital ratios, stricter lending standards), historical parameters don't apply.

****How to Address:**** Subsample analysis (pre-2008 vs. post-2009) would isolate regime-specific parameters. We did conduct this via robustness testing in M3 and found leverage effect magnitudes differed significantly across periods, confirming this concern is justified. This limitation supports our memo's caution on leverage interpretation.

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****Task 1: DiD parallel-trends visual inspection****

- ****Prompt:**** "Create a visualization showing pre-2015 return trends for large-cap and small-cap REITs separately to assess the parallel trends assumption."
- ****Output:**** AI generated code to plot two time series (large-cap avg return, small-cap avg return) for 2000–2015 with legend and labels.
- ****Verification:**** Ran code; visually inspected plot. Trends appear roughly parallel (correlation ≈ 0.92 over 2000–2015), supporting validity of DiD. Found one brief divergence 2011–2012 (small-cap outperformance), possibly correlated with Fed tapering expectations, which AI didn't flag.
- ****Changes:**** Added text note about 2011–2012 divergence and sensitivity of DiD to this period. Acknowledged potential violation of strict parallel trends but defended estimate as approximate.

****Task 2: Sector-level return calculations****

- ****Prompt:**** "Calculate average monthly returns by REIT sector (Industrial, Retail, Office, Residential) over 2000–2024 and

test for statistical differences using ANOVA." - **Output:** AI provided correct ANOVA syntax with p-values and pairwise comparisons. - **Verification:** Results: F-stat = 8.73, $p < 0.001$. Industrial REITs avg return = 1.14%/month; Office = 0.81%/month. Differences significant. Spot-checked calculations manually for two sectors. - **Changes:** None; AI output was accurate and directly fed into sector recommendation (Overweight Industrial based on empirical returns differential).

Task 3: Null result interpretation writing - **Prompt:** "Write an interpretation of a null DiD finding (coefficient $\approx$ zero, $p > 0.05$) that explains why the null is informative rather than disappointing." - **Output:** AI provided template explaining that null findings rule out certain hypotheses and can strengthen confidence in alternative theories. - **Verification:** Reviewed for tone and accuracy; AI template was appropriate but lacked our specific context. Tailored to DiD null finding: "suggests large-cap hedging was effective OR size is not the binding constraint on rate exposure." - **Changes:** Added specific business interpretation: null result points toward beta (systematic risk) as the true driver rather than leverage, aligning with our investment recommendations. **Overall Assessment:** AI provided technical scaffolds and standard statistical interpretation templates. All numerical results derived from running statistical software on actual M1-M3 data. Interpretation added human judgment and business context not provided by AI. --- **Status:** ✓ COMPLETE **Submission Date:** May 1, 2026 by 11:59 PM EST